CRAMÉR-RAO BOUNDS ON SPARSE-DIFFUSE CHANNEL ESTIMATION

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ABSTRACT

Lower bounds on the estimation of a hybrid sparse-diffuse channel and parameters of the channel are provided. The Hybrid Atomic-Least-Squares (HALS) algorithm was designed to jointly estimate the sparse and diffuse components with a combined atomic and ℓ_2 regularization for this hybrid, mixed-channel model. The Cramér-Rao Bound analysis in this paper focuses on the estimation of the channel parameters, resulting in a bound on the aggregate channel. Numerical results via simulations on synthetic data validate the efficacy of the HALS estimation strategy and show the improved predictive ability of the CRB analysis for the performance of HALS versus previously considered bounds.

Index Terms— Channel estimation, convex optimization, atomic norm minimization, source separation, Cramér-Rao bound.

1. INTRODUCTION

Most applications within wireless communications rely on high-performance and reliable channel estimation. In many scenarios, such as free-space propagation environments, wireless communication channels can be modeled as finite impulse response filters with a sparse number of taps. In these cases, sparse approximation methods yield high performance (*e.g.* [1, 2]). In other environments, such as indoor, a mixed model of diffuse and specular, sparse components has been proposed in [3]. The model versatility is shown to result in better channel fits [4, 5, 6]. Yet, the proposed estimators rely on frequency discretization, inducing a potential reconstruction mismatch.

In [7], an *Hybrid Atomic Least-Squares* (HALS) algorithm was proposed for high-performance estimation of hybrid sparse-diffuse channels (HSD). HALS is adapted from the atomic norm denoiser [8, 9] to harness the deterministic structure of the diffuse components, while estimating the sparse components without relying on discretization.

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Deriving a meaningful Cramér-Rao bound (CRB) for the HSD channel model is challenging, as the involved parameters act differently on the observation, and since the sparse model order is unknown. Prior work [7] adapted results on constrained estimators to propose two approximate CRBs [10, 11] by treating one of the sparse/diffuse components as noise; however, these approximate CRBs are loose.

The main contribution of this paper is a derivation of a CRB for the entire HSD channel, which includes *both* sparse plus diffuse components from the perspective of fitting a model [12]. In Proposition 1, the CRB on the channel parameters is related to two quantities that are dependent on the separation between the sparse and diffuse frequencies, and independent of the model order. Our proof technique harnesses recent advances in the condition number of confluent Vandermonde matrices [13]. Then, a CRB on the channel coefficient, transformed from the CRB on channel parameters [14], is provided in Proposition 2.

This paper is organized as follows. Section 2 describes the sparse-diffuse channel model. Section 3 reviews the HALS estimator [7]. The new CRB derivation is presented in Section 4. Numerical results are provided in Section 5. The new CRB is seen to be predictive of HALS performance.

2. SIGNAL AND CHANNEL MODEL

For brevity, we directly provide the discrete-time equivalent channel model, which, assuming an OFDM signaling, results from sampling the output of a matched filter with sampling period T_s. The discrete signal received over the k-th subcarrier is denoted by y_k. Stacking the subcarriers signals into a vector $\boldsymbol{y} = [y_0, \cdots, y_{N-1}]^\top \in \mathbb{C}^N$, one may write

$$\boldsymbol{y} = \boldsymbol{S}\boldsymbol{h} + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{S} = \operatorname{diag}(s_0, s_1, \cdots, s_{N-1})$ is the pilot matrix, $\boldsymbol{h} = [h_0, h_1, \cdots, h_{N-1}]^\top$ is the channel coefficients vector, and $\boldsymbol{n} = [n_0, n_1, \cdots, n_{N-1}]^\top$ is additive noise.

The sparse-diffuse channel is further modeled as $\boldsymbol{h} = \boldsymbol{h}_s + \boldsymbol{h}_d$, where $\boldsymbol{h}_s$ and $\boldsymbol{h}_d$ are the sparse component and diffuse components of the channel, respectively. As the presence of multipath causes distinct delays in the time domain, $\boldsymbol{h}_s$ can be assumed to be a sparse linear combination of com-

plex exponential vectors of the form $\boldsymbol{a}(f)$, defined by

$$\boldsymbol{a}(f) \triangleq \left[1, e^{j2\pi f}, \cdots, e^{j(N-1)2\pi f}\right]^{\top}, \quad \forall f \in [0,1). \quad (2)$$

Hence, assuming m sparse components, the sparse contribution in the HSD channel model can be expressed as

$$\boldsymbol{h}_s = \sum_{i=1}^{m} \alpha_i \boldsymbol{a}\left(-\frac{\tau_i}{T_s}\right) = \sum_{i=1}^{m} \alpha_i \boldsymbol{a}(f_i) = \boldsymbol{A_f}\boldsymbol{\alpha}, \quad (3)$$

where τ_i is the path delay, $f_i \triangleq \frac{T_s-\tau_i}{T_s} \in [0,1)$, $\boldsymbol{A_f} = [\boldsymbol{a}(f_1), \cdots, \boldsymbol{a}(f_m)] \in \mathbb{C}^{N\times m}$, and $\boldsymbol{\alpha} = [\alpha_1, \cdots, \alpha_m] \in \mathbb{C}^m$ is the sparse amplitude vector. Under the assumption that $m \ll N$, the component $\boldsymbol{h}_s$ in (3) admits a sparse decomposition over the continuous atomic dictionary $\{\boldsymbol{a}(f) : f \in [0,1)\}$.

Similarly, the diffuse components are modeled as a linear combination of atoms of the form (2). Yet, unlike the sparse components, frequencies are *fixed and known*, and assumed to lie on the discrete grid $\{0, \frac{1}{L}, \cdots, \frac{L-1}{L}\}$ for some $L < N$. Hence, we can write

$$\boldsymbol{h}_d = \sum_{r=1}^{L} \gamma_r \boldsymbol{a}\left(\frac{r-1}{L}\right) = \boldsymbol{D}\boldsymbol{\gamma}, \quad (4)$$

where $\boldsymbol{\gamma} = [\gamma_1, \cdots, \gamma_L]^{\top} \in \mathbb{C}^L$ is the diffuse amplitude vector, and $\boldsymbol{D} = \left[\boldsymbol{a}(0), \boldsymbol{a}\left(\frac{1}{L}\right), \cdots, \boldsymbol{a}\left(\frac{L-1}{L}\right)\right] \in \mathbb{C}^{N\times L}$ is the dictionary matrix for the diffuse components. We underscore that the diffuse components here model an aggregate of diffuse components at arbitrary delays. In general, one could employ a single model with all paths at arbitrary delays; however, doing so would not effectively capture the nature of the channel, which includes both strong specular and diffuse components. Furthermore, the former approach would yield a channel that is not sparse, thus limiting the gains typically afforded by sparse methods.

In practical applications, the diffuse amplitudes are often governed by an exponential damping factor $|\gamma_r|^2 \approx Ce^{-\beta r}$. We also assume that the energy of the diffuse coefficients $\|\boldsymbol{\gamma}\|_2^2$ is small relative to that of the specular multipath $\|\boldsymbol{\alpha}\|_2^2$ [7].

Finally, with (1), (3) and (4) the overall HSD channel estimation problem is to estimate $\{\hat{\boldsymbol{f}}, \hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\gamma}}\}$ given the observation model

$$\boldsymbol{y} = \boldsymbol{S}\left(\boldsymbol{A_f}\boldsymbol{\alpha} + \boldsymbol{D}\boldsymbol{\gamma}\right) + \boldsymbol{n}. \quad (5)$$

3. HYBRID ATOMIC-LEAST-SQUARES

The HALS estimator was introduced to estimate the HSD channel in [7] as the solution of the convex optimization program

$$\arg\min_{\boldsymbol{h}_s,\boldsymbol{\gamma}} \frac{1}{2}\left\|\boldsymbol{S}^{-1}\boldsymbol{y} - \boldsymbol{h}_s - \boldsymbol{D}\boldsymbol{\gamma}\right\|_2^2 + \tau\|\boldsymbol{h}_s\|_{\mathcal{A}} + \frac{\lambda}{2}\|\boldsymbol{\gamma}\|_2^2, \quad (P1)$$

where $\tau \geq 0$ and $\lambda \geq 0$ are the hyper-parameters that need to be determined. Intuitively, (P1) is the minimization of three convex terms: 1) a model fidelity term; 2) an atomic norm minimization term, promoting the sparsity of $\boldsymbol{h}_s$ into the atoms (2); 3) a Tikhonov regularization term on $\boldsymbol{\gamma}$ promoting low-energy on the diffuse component, and enhancing stability. By leveraging semidefinite representation of the atomic norm [15] and properties of Lagrangian multipliers, it is shown in [7] that the HALS program (P1) can be reformulated as a semidefinite program (SDP)

$$\min_{t,\boldsymbol{\iota},\boldsymbol{h}_s,\boldsymbol{\gamma}} \left\|\boldsymbol{S}^{-1}\boldsymbol{y} - \boldsymbol{h}_s - \boldsymbol{D}\boldsymbol{\gamma}\right\|_2^2 + \tau(t+\iota_1)$$
$$\text{s.t.} \begin{bmatrix} \text{Toep}(\boldsymbol{\iota}) & \boldsymbol{h}_s \\ \boldsymbol{h}_s^{\mathsf{H}} & t \end{bmatrix} \succeq 0 \text{ and } \|\boldsymbol{\gamma}\|_2^2 \leq E_d, \quad (P2)$$

where ι_1 is the first entry of $\boldsymbol{\iota}$, $\text{Toep}(\boldsymbol{\iota})$ is the Hermitian Toeplitz matrix whose first column is the vector $\boldsymbol{\iota}$, and E_d is an upper bound on the energy estimate on the diffuse components' amplitude. The HALS estimator can directly be calculated by solving (P2) using off-the-shelf convex solvers such as CVX [16, 17]. We denote the optimal solution as $\hat{\boldsymbol{h}}_s$ and $\hat{\boldsymbol{\gamma}}$. Then, the frequency support of the sparse components can be inferred as [7, 18]

$$\widehat{\mathcal{T}} = \left\{\hat{f} \in [0,1) : \left|\left\langle \boldsymbol{S}^{-1}\boldsymbol{y} - \hat{\boldsymbol{h}}_s - \boldsymbol{D}\hat{\boldsymbol{\gamma}}, \boldsymbol{a}\left(\hat{f}\right)\right\rangle\right| = \tau\right\}. \quad (6)$$

Once the frequency support $\widehat{\mathcal{T}}$ has been estimated with (6), we denote by $\boldsymbol{P}_{\widehat{\mathcal{T}}} \in \mathbb{C}^{N\times N}$ the orthogonal projection matrix unto the span of $\left\{\boldsymbol{a}\left(\hat{f}\right), \forall \hat{f} \in \widehat{\mathcal{T}}\right\}$, then the de-biased estimate [8] of the resolvable paths is given by

$$\hat{\boldsymbol{h}}_{s,\text{db}} = \boldsymbol{P}_{\widehat{\mathcal{T}}}\left(\boldsymbol{S}^{-1}\boldsymbol{y} - \boldsymbol{D}\hat{\boldsymbol{\gamma}}\right), \quad (7)$$

which is equivalent to projecting the estimated residual sparse component onto the span of the dictionary generated by the atoms of the estimated sparse support.

We observe that our model generalizes the classical, purely sparse approach, by setting $\boldsymbol{\gamma} = \boldsymbol{0}$ in (P1) and (P2), we recover the original atomic norm denoising method [8]. We denote this solution as $\hat{\boldsymbol{h}}_{s,\text{ANM}}$.

4. ANALYSIS OF THE CRAMÉR-RAO BOUND

We wish to understand the fundamental limits of channel estimation given the channel description in Section 2. In this section, we present our main contribution, which is a tight characterization of the Cramér-Rao lower bound (CRB) for estimating both the channel parameters and the channel coefficients.

4.1. Cramér-Rao bound on the channel parameters

We start in this section by deriving the Fisher information matrix (FIM) for the ground truth HSD channel parameters

$\{\boldsymbol{f}, \boldsymbol{\alpha}, \boldsymbol{\gamma}\}$ given the observation model of Equation (5). For simplicity, we assume all the parameters are complex.

Lemma 1. *Assume $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$, and denote the parameter vector $\boldsymbol{\theta} = \left[\boldsymbol{f}^\top, \boldsymbol{\alpha}^\top, \boldsymbol{\gamma}^\top\right]^\top$. Then, the Fisher information matrix $\boldsymbol{J}^{\boldsymbol{\theta}}$ on $\boldsymbol{\theta}$ given the measurement model* (5) *is given by*

$$\boldsymbol{J}^{\boldsymbol{\theta}} = \sigma^{-2} \left[\boldsymbol{\Lambda} \boldsymbol{A}_{\boldsymbol{f}} \operatorname{diag}(\boldsymbol{\alpha}), \boldsymbol{A}_{\boldsymbol{f}}, \boldsymbol{D}\right]^{\mathsf{H}} \boldsymbol{S}^{\mathsf{H}} \boldsymbol{S} \left[\boldsymbol{\Lambda} \boldsymbol{A}_{\boldsymbol{f}} \operatorname{diag}(\boldsymbol{\alpha}), \boldsymbol{A}_{\boldsymbol{f}}, \boldsymbol{D}\right] \tag{8}$$

where $\boldsymbol{\Lambda} = \operatorname{diag}(0, j2\pi, \cdots, j2\pi(N-1)) \in \mathbb{C}^{N \times N}$.

Proof. Since $\boldsymbol{y} \sim \mathcal{CN}\left(\boldsymbol{S}\boldsymbol{h}, \sigma^2 \boldsymbol{I}_N\right)$ by the assumption, the FIM on $\boldsymbol{h}$ given $\boldsymbol{y}$ is given by $\boldsymbol{J}^{\boldsymbol{h}} = \sigma^{-2} \boldsymbol{S}^{\mathsf{H}} \boldsymbol{S}$. Moreover, from the chain rule, one has $\boldsymbol{J}^{\boldsymbol{\theta}} = \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{\theta}}\right)^{\mathsf{H}} \boldsymbol{J}^{\boldsymbol{h}} \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{\theta}}\right)$. It is enough to notice by a direct calculation that

$$\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{\theta}} = \left[\boldsymbol{\Lambda} \boldsymbol{A}_{\boldsymbol{f}} \operatorname{diag}(\boldsymbol{\alpha}), \boldsymbol{A}_{\boldsymbol{f}}, \boldsymbol{D}\right] \tag{9}$$

to conclude on the desired statement. □

The Cramér-Rao lower bound states that the covariance $\boldsymbol{K}$ of any unbiased estimator satisfies $\boldsymbol{K} \succeq \left(\boldsymbol{J}^{\boldsymbol{\theta}}\right)^{-1}$ for the semi-definite order. Therefore, the crux of the problem is to control the inverse of (8). In [13], a stability analysis of the super-resolution problem is proposed by characterizing the extremal eigenvalues of the CRB of the problem. Herein, we harness the Vandermonde structure of matrices $\boldsymbol{\Lambda} \boldsymbol{A}_{\boldsymbol{f}}$, $\boldsymbol{A}_{\boldsymbol{f}}$ and $\boldsymbol{D}$ in the expression of the FIM (8), analogously to that of super-resolution settings, to propose novel statistical bounds on the CRB of the HSD channel parameters.

A key metric that measures problem stability is the *minimum separation* between the support of the sparse and diffuse components [13, 19], which we define as the minimum wrap-around distance between the sparse support by $\delta_{\mathrm{s}} = \min_{i \neq j} \min_{k \in \mathbb{Z}} \{|f_i - f_j + k|\}$, and the distance between the sparse and diffuse support by $\delta_{\mathrm{s-d}} = \min_i \min_{r \in \mathbb{Z}} \{|f_i - \frac{r}{L}|\}$. Furthermore, for clarity, the pilot matrix is assumed unitary, *i.e.* $\boldsymbol{S}^{\mathsf{H}} \boldsymbol{S} = \boldsymbol{I}_N$. In practice, this can be achieved by transmitting phase-shift keying (PSK) data symbols with unit energy.

Proposition 1. *Assume $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$ and $\boldsymbol{S}^{\mathsf{H}} \boldsymbol{S} = \boldsymbol{I}_N$. Let $\delta = \min(\delta_{\mathrm{s}}, \delta_{\mathrm{s-d}})$. Furthermore, assume $\alpha_i \neq 0$ for all $i \in [\![1, m]\!]$. If $2m + L \leq N$ and $N\delta > 2$, then*

$$\begin{aligned} K_{\min} |\alpha_i|^{-2} \sigma^2 &\leq \frac{2\pi^2 (N-1)(2N-1)}{3} \mathrm{CRB}(f_i) \\ &\leq K_{\max} |\alpha_i|^{-2} \sigma^2, \end{aligned} \tag{10a}$$

$$K_{\min} \frac{\sigma^2}{N} \leq \mathrm{CRB}(\alpha_i) \leq K_{\max} \frac{\sigma^2}{N}, \tag{10b}$$

$$K_{\min} \frac{\sigma^2}{4N} \leq \mathrm{CRB}(\gamma_r) \leq K_{\max} \frac{\sigma^2}{4N}, \tag{10c}$$

for all $i \in [\![1, m]\!]$ and all $r \in [\![1, L]\!]$, where $K_{\min}$ and $K_{\max}$ are the constants depending on the separation parameter $N\delta$

$$K_{\min} = \frac{4}{\left(1 + 2(N\delta)^{-1}\right)}, \tag{11}$$

$$K_{\max} = \frac{4}{\left(1 - 2(N\delta)^{-1}\right)}. \tag{12}$$

Proposition 1 provides the CRB on each individual parameter of the HSD channel. The proposed bound indicates that the CRB on f_i (10a) decays as N^{-1} when the separation parameter $N\delta = \mathcal{O}(1)$ is fixed, which corroborates with the empirical Rayleigh limit, predicting a maximum resolution power of the order $\delta \simeq N^{-1}$ [20]. Similarly, the CRBs (10b) and (10c) decays as N^{-1} under a fixed noise level σ. Second, it is interesting to pinpoint that the bounds (10) are independent of the model orders m and L. Finally, our lower and upper estimates of the CRBs become tighter for a large separation parameter $N\delta$, providing a sharper estimate. Proposition 1 yields a significant improvement to [7], where approximate CRBs were derived alternatively for the sparse and the diffuse channel by modeling the other channel as Gaussian noise.

4.2. Cramér-Rao bound on the HSD channel

Proposition 2. *Assume $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$ and $\boldsymbol{S}^{\mathsf{H}} \boldsymbol{S} = \boldsymbol{I}_N$. If $2m + L \leq N$ then any unbiased estimator $\hat{\boldsymbol{h}}$ of $\boldsymbol{h}$ satisfies*

$$\mathbb{E}\left[\left(\boldsymbol{h} - \hat{\boldsymbol{h}}\right)\left(\boldsymbol{h} - \hat{\boldsymbol{h}}\right)^{\mathsf{H}}\right] \succeq \mathrm{CRB}(\boldsymbol{h}) = \sigma^2 \boldsymbol{P}_{\boldsymbol{\theta}}, \tag{13}$$

where $\boldsymbol{P}_{\boldsymbol{\theta}}$ is the orthogonal projector unto the span of $[\boldsymbol{\Lambda} \boldsymbol{A}_{\boldsymbol{f}} \operatorname{diag}(\boldsymbol{\alpha}), \boldsymbol{A}_{\boldsymbol{f}}, \boldsymbol{D}]$.

Proof. The result is immediate from (9), Lemma 1 and the CRB for the forward transformation of parameters [12, 14] $\mathrm{CRB}(\boldsymbol{h}) = \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{\theta}}\right) \left(\boldsymbol{J}^{\boldsymbol{\theta}}\right)^{-1} \left(\frac{\partial \boldsymbol{h}}{\partial \boldsymbol{\theta}}\right)^{\mathsf{H}}$. □

A direct consequence of Proposition 2 is that the mean-squared error of an unbiased estimator of the channel $\boldsymbol{h}$ is lower-bounded by $\mathbb{E}\left[\left\|\boldsymbol{h} - \hat{\boldsymbol{h}}\right\|_2^2\right] \geq \operatorname{tr}(\mathrm{CRB}(\boldsymbol{h})) = \sigma^2 (2m + L)$, which is proportional to the number of unknown parameters.

5. NUMERICAL RESULTS

Our experimental setup is as follows. Rayleigh fading [3] is assumed for the path gains, so that $\alpha_i \sim \mathcal{CN}\left(0, e^{-\omega \frac{\tau_i}{\Delta T}}\right)$ for $i = [\![1, m]\!]$, and the $\gamma_r \sim \mathcal{CN}\left(0, \beta e^{-\omega r}\right)$ for $r = [\![0, L-1]\!]$. The noise is drawn from $\boldsymbol{n} \sim \mathcal{CN}\left(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N\right)$. Our data symbols are drawn uniformly from the QPSK constellation. We assume that $\boldsymbol{D}$ is known. E_d is chosen to take

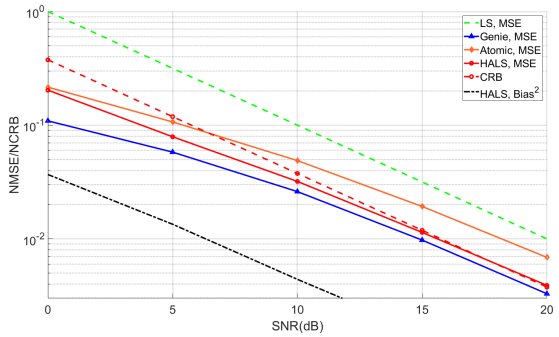


Fig. 1. Normalized MSE/CRB on the channel coefficients against SNR. Parameter values are: $N = 101$, $L = 30$, $\omega = 0.05$, $\beta = 0.01$, $m = 4$ and $\delta = 1/(2L)$.

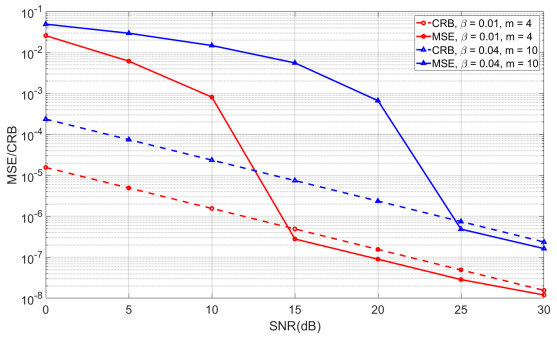


Fig. 2. MSE/CRB on frequency estimation against SNR. Parameter values are: $N = 101$, $L = 30$, $\omega = 0.05$ and $\delta = 1/(2L)$.

the value of the expected amplitude of $\boldsymbol{c}_d$. We define the signal-to-noise ratio SNR(dB) $= 10\log_{10}\frac{\|\boldsymbol{h}^\star\|_2^2}{N\sigma^2}$. Our performance metric is the normalized mean square error (NMSE): NMSE $= \frac{\|\boldsymbol{h}^\star - \hat{\boldsymbol{h}}\|_2^2}{\|\boldsymbol{h}^\star\|_2^2}$, where $\boldsymbol{h}^\star$ is the true channel and $\hat{\boldsymbol{h}}$ is the estimated channel. For our HALS algorithm, $\hat{\boldsymbol{h}}_{\text{HALS}} = \hat{\boldsymbol{h}}_{s,\text{db}} + \boldsymbol{D}\hat{\boldsymbol{\gamma}}$. Here, the hyperparameters for both HALS and vanilla ANM are fine-tuned to give an estimate with the correct number of sparse components and to yield the best MSE performance as in [21]. As a reference, the genie-aided scheme [7], which knows the true support of sparse components, is implemented. The genie provides an estimate $\hat{\boldsymbol{h}}_{\text{ge}} = \hat{\boldsymbol{h}}_{s,\text{ge}} + \hat{\boldsymbol{h}}_{d,\text{ge}}$. The least-squares estimator of the channel is given by $\hat{\boldsymbol{h}}_{\text{LS}} = \boldsymbol{S}^{-1}\boldsymbol{y}$. The convex problems are solved with CVX [16, 17]. For consistency with the NMSE, the CRBs are normalized by $\|\boldsymbol{h}^\star\|_2^2$. The results are averaged over 20 synthetic channel realizations.

Figures 1 benchmarks the performance of the estimators to recover the channel coefficient $\boldsymbol{h}$ against the CRB, when the number of unknown parameters $2m + L$ is 38, and the relative energy parameter β is 0.01. Clearly, HALS improves channel estimation compared to the ANM estimator, particularly in the high-SNR region. We underline that the performance of HALS approaches the CRB in the high-SNR regime. As the HALS estimator (P2) is *biased* due to the regularization, its performance can be locally lower than CRB.

Figure 2 and 3 show the performance of HALS in estimating the frequencies of sparse components against the CRB, under varying SNR and minimum separation values δ, respectively. Two sparsity and energy levels: $(m = 4, \beta = 0.01)$ and $(m = 10, \beta = 0.04)$, are considered. In both cases, the biased estimator HALS performs better than the CRB in the high-SNR regime and for channels with sparse and diffuse components separated enough, supporting the efficacy of our estimator. We see the breakdown points in both experiments. For channels that better fit our model, the large performance improvement occurs at smaller SNR and minimum separation values. Furthermore, the CRB values obtained at $N\delta = 2.525 > 2$ in Figure 3 are compared to the upper and lower bounds on the CRB derived in Proposition 1, validating the theorem statement.

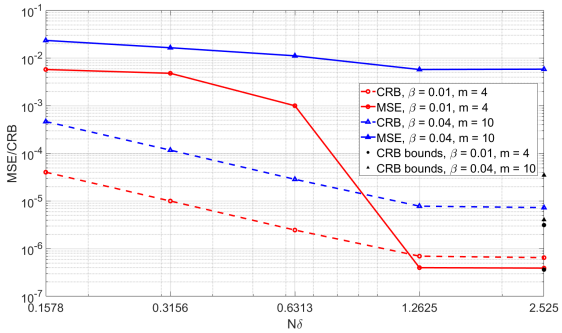


Fig. 3. MSE/CRB on frequency estimation against the separation parameter $N\delta$. Parameter values are: $N = 101$, $L = 20$, $\omega = 0.05$ and SNR $= 15$dB.

6. CONCLUSIONS

In this work, we derive a CRB for parameter estimation in our HSD channel model and propose upper and lower bounds for it. The bounds on the CRB depend on the channel's structure, including the minimum separation between spikes, the minimum and maximum spike amplitudes, the channel length, and the noise magnitude. We review the properties of the HALS estimator for sparse-diffuse channel estimation, and show our newly derived CRBs tightly explain HALS performance, especially in the high-SNR regime. The results constitute a major improvement over those previously derived in [7]. Furthermore, our numerical results show the improved performance of the HALS estimator over the vanilla ANM estimator, both in terms of channel estimation and in the recovery of sparse components, which was anticipated as HALS harnesses the full structure of the HSD channel.

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